

Four drones are assigned to three urban pollution sensing zones. Task 1 requires two drones, whereas Tasks 0 and 2 each require a single drone. The goal is to minimize the sum of Euclidean distances for all assigned drone-task pairs. The positions of the drones are represented in matrix: p_i and the task positions p_j .

Natural-language task Specifications(u)



Parser Agent
($\mathcal{A}_{Par}$)

Given task input u &
Memory $\mathcal{M}$

Assignment matrix:

$x_{ij} \in \{0, 1\}$, where $x_{ij} = 1$
denotes drone i is assigned to task j .

Drones coordinate:

$p_i = (x_i, y_i)$,

Tasks coordinate :

$p_j = (x_j, y_j)$,

Requirements:

$r_0 = 1, r_1 = 2, r_2 = 1$
denotes task j require k drones



Retrievable Memory Repository ($\mathcal{M}$)



Modeler Agent
($\mathcal{A}_{Mod}$)

Given representation z
& Memory $\mathcal{M}$

Assignment matrix:

$x_{ij} \in \{0, 1\}$ for $i \in \{0..3\}, j \in \{0..2\}$

Euclidean distance:

$d_{ij} = ||p_{di} - p_{tj}||$

Constraints:

$$\sum_{j=0}^2 x_{ij} = 1, \sum_{j=0}^2 x_{i0} = 1,$$

$$\sum_{j=0}^2 x_{i1} = 2, \sum_{j=0}^2 x_{i2} = 1, \forall i = 0 \dots 3$$

Objective:

$$\min \sum_{i=0}^3 \sum_{j=0}^2 d_{ij} \cdot x_{ij}$$



Developer Agent
($\mathcal{A}_{Dev}$)

Given optimization
problem Π & Memory $\mathcal{M}$

```
import gurobipy as gp
import numpy as np
```

```
robot = np.array(pos)
model=gp.Model("RobotTask
")
...
model.optimize()
return assignment
```

System Output



**Executable
Solver Code(c)**

**Executable
Solver Code
Program Output:**
[0,1,2,1]
**Ground
Truth:[0,1,2,1]**

